



An estimation of physicochemical properties of heart attack treatment medicines by using molecular descriptor's

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ABSTRACT

A topological index is a numerical quantity associate with a graph describes its topology by applying molecular structure of a chemical compound, we obtain its properties theoretically. The chemical structures of anti-heart attack drugs are investigated and analyzed by using the several degree-based topological indices. Heart attacks, also known as myocardial infarctions, are extremely prevalent in the world. In the QSAR model, degree-based topological indices computed through the edge partitioning approach, while correlation coefficients determined by a linear regression model and line graphs used to compare the correlations. The statistical analysis of various characteristics including boiling point, vapor pressure, enthalpy, flash point, molar refraction, surface tension, and molar volume in anti-heart attack medicines, suggests a strong correlation with topological indices which signify the importance of all computations. All positive correlation values indicate a direct correlation to the molecular descriptors. The characteristics of the anti-heart attack medicines computed by our simple mathematical method are better as compare to experimental values of the heart attack drugs obtained with time-consuming and expensive laboratory experiments.

1. Introduction

Myocardial infarction, another name for a heart attack, occurs when the blood supply to a section of the heart muscle is insufficient. The affected portion of the heart may get damaged if there is insufficient oxygen-rich blood circulating there. A blockage in the arteries carrying blood to your heart is often what causes a heart attack. A higher risk of heart failure and other severe consequences can result from the heart not receiving the blood and oxygen it needs to work efficiently. The whole heart's pumping sequence is destroyed when a portion of the heart cannot pump because of a shortage of blood (Merritt et al. (2017); National Heart et al. (2016)).

There are several possible symptoms of a heart attack, such as chest pain, feeling weak or lightheaded, pain in the neck, jaw, back, or shoulder, pain in one or both arms, shortness of breath, nausea, sweating, fatigue, upper body pain, and trouble breathing. These symptoms vary according to gender. There are two major types of heart attacks. The plaque on the inner wall of the artery ruptures, and cholesterol and other chemicals are released into the circulation, resulting in type I heart attacks. Type II heart attacks do not completely block an artery, but the heart does not receive the amount of oxygen-rich blood that it requires.

Smoking, high cholesterol, obesity, lack of exercise, high levels of stress, diabetes or prediabetes, eating a diet heavy in trans and saturated fats, excessive alcohol usage, and sleep apnea are the leading causes of heart attacks (KRUEGER et al. (1981)).

A heart attack is a medical emergency that poses a risk to life. Every year, 17 million people die from heart disease throughout the world. In Pakistan, heart-related diseases were responsible for about 19 percent of fatalities in 2020; this percentage has since increased to 29 percent. Some procedures to prevent and treat a heart attack are stent placement, angioplasty, heart bypass surgery, heart valve surgery, a heart transplant, etc. (Davis et al. (1996)). The production of medicine for the treatment of this disorder is very expensive, time-consuming, and difficult. So chemists defined different tools to easily understand these molecular structures.

A molecular graph is a graph-theoretical model of the structural formula of a chemical compound, where the vertices represent the complex's atoms and the edges represent its chemical bonds. A matrix, a polynomial, a series of integers, or a numeric number can all be used to represent a graph. A new field of study called "chem-informatics" combines chemistry, information science, and math. It investigates correlations between quantitative structure-activity (QSAR) and

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structure-property (QSPR) that are used to estimate the biological activities and characteristics of various chemical compounds. A topological index, which characterizes a graph's topology and is invariant under graph automorphism, is a numerical value related to the graph. Topological indices are unique among graph invariants and are used to predict the physio-chemical properties of chemical compounds (Dearden (2017)).

A T-index is a function $h: G \rightarrow R$ that converts each graph of a chemical structure into a numerical value. It differs from other graph invariants in that it can calculate the physicochemical properties of a chemical molecule. It separates from other graph invariants because it is able to calculate the physio-chemical features of a chemical molecule. There are many classifications of indices, such as vertex degree based, distance-based, eccentricity-based, ev-degree-based, etc. The concept of indices is given by Harold Wiener (Wiener (1947)) while estimating the boiling point of organic compounds. He called his index a path number. Milan Randić proposed the most common degree-based index, known as the Randić index, in 1975 (Randić (1975)).

This article discusses the linear regression model and degree-based indices. There are numerous articles with applications for degree-based indices. Rauf et al. applied the eigen values approach for the analysis of medicines in COVID-19 (Rauf et al. (2022a)), the ev-degree and ve-degree method for understanding the breast cancer drugs (Rauf et al. (2022b)), entropy-based indices for the computation of the properties of benzene derivatives (Rauf et al. (2022a)), and the ev-degree and ve-degree approach for the analysis of benzene derivatives (Rauf et al. (2022a)). Adnan et al. (Adnan et al. (2022)) use a linear regression model and 11 degree-based topological indices in their research investigation on anti-tuberculosis drugs. He noted that the Randić index is used to estimate molar refraction ($r = 0.984$), the geometric arithmetic index is useful for calculating molar volume ($r = 0.917$), the harmonic index can be used to calculate enthalpy ($r = 0.891$), and the sum connectivity index is used to determine boiling point ($r = 0.919$). The ABC-index has the highest correlation with polarization ability ($r = 0.980$).

Recently, Zhang (Zhang et al. (2023)) applied degree-based indices and statistical methodology to anti-malarial medications. Using the same method as in this manuscript, the characteristics of the anti-malarial medications are estimated. He observed that the first Zagreb index has a 0.963 relationship with enthalpy. The harmonic index has a good correlation with polarizability, the forgotten index with enthalpy has $r = 0.969$, the SS index has the highest correlation with boiling point and flash point, both of which are 0.948, and the redefined Zagreb indices with enthalpy have 0.947, 0.958, and 0.962, respectively. For the computation, analysis, and investigation of the two-dimensional structures of NSAID medicines, Gnanaraj et al. utilized the degree-based indices (Gnanaraj et al. (2023)). In 2023, Azeem et al. studied the characteristics of anxiety drugs with the help of topological indices and a linear regression model (Azeem and Jamil (2023)).

The contribution of our work to this paper is described in the following points:

- The results are calculated using the ten topological indices. There are several types of molecular descriptors, particularly topological indices, although classical degree-based indices are appropriate for investigating heart attack medications.
- The linear regression model is used to determine an approximate linear relationship between topological indices and experimental graph data. Simple linear regression is used to investigate the relationship between two variables. Usually, the goal is to calculate the value of an output variable based on the value of an input variable.
- The values of the physicochemical properties have already been derived using lab tests, but we use a statistical technique to estimate the properties of the heart attack. All regression equations and statistical parameters are computed using SPSS software.

- The correlation coefficient, whose values represent the validity of the theoretical experiments, is the most important aspect of the calculations. These parameters can help chemists and pharmacists develop drugs for heart attack disease. The p-test shows the significance of the correlation coefficients. For significant results, the correlation coefficients must be high and the p-value must be low.
- All of the lines in the graphs are drawn with the use of Microsoft Excel for simple analysis of the results, which are presented graphically.

The paper is organized as follows: Section 2 defines all topological indices and provides a brief literature review of all indices. Section 3 explains all of the techniques used to compute topological index-related results. Tables of experimental values and topological indices of heart attack medications are included in Section 4. Section 5 contains all statistical computations. Section 6 provides a graphical comparison of all the correlation coefficients. Section 7 discusses the applications of the topological indices used in the article, and Section 8 contains the results discussion and concluding remarks. References are listed at the end of the paper.

2. Basic definitions and literature review

A simple graph $G(V, E)$ with a collection of nodes $V(G)$ and a collection of edges $E(G)$ is connected if any pair of nodes in G has a connection. All the graphs used in this article are 2D, simple (without multiple edges and loops), undirected (unordered), and planar (without any crossings). The symbol $d(r, s)$ represents the shortest distance between any two vertices r and s . The number of edges incident on any vertex r is called its degree and is denoted by Φ_r . The maximum vertex degree is three, and the minimum vertex degree is one. An edge $e = rs = (r, s)$ is formed by the connection of two vertices, or it is a bridge between two vertices. The idea of degree is linked to the concept of valence in chemistry. For more information about the basics of graph theory, see (Sonawane et al. (2022)).

1. Randić Index

Milan Randić, a chemist, introduced a topological index called the "branching index" in 1975 to measure the level of branching in the carbon-atom skeleton of organic compounds (Randić (1975)).

$$R(G) = \sum_{r,s \in E(G)} \frac{1}{\sqrt{\Phi_r \times \Phi_s}}$$

For the enthalpy, flash point, molar refraction, and polarizability of breast cancer medicines, the Randić index demonstrates strong positive and very significant correlation values of $r = 0.977$, 0.872 , 0.987 , and 0.986 , respectively (Bokhary et al. (2021)). Amylose and blue starch iodine are the complex molecules; different forms of the R-index are used to understand the properties of these molecules (Rani and Ali (2021)).

2. Reciprocal Randić Index

Gutman proposed the reciprocal Randić index in 2014 (Gutman et al. (2014)). The mathematical statement of the RR-index is:

$$RR(G) = \sum_{r,s \in E(G)} \sqrt{\Phi_r \times \Phi_s}.$$

3. Harmonic Index

Fajtlowicz first suggested the harmonic index in 1987 (Fajtlowicz (1987)). He suggested this index for the computer networks and was a computer engineer. For chemical graphs, especially trees, Zhong independently suggested a similar degree-related index in 2012 (Zhong (2012)). The H-index is defined as follows:

Table 1
Physicochemical characteristics of heart attack drugs

Drug Name	BP	VP	EV	FP	MR	ST	MV
Apixaban	770.5±60	0.0±2.6	112.2	419.8±3	125.6	61.0	323.4
Aspirin	321.4±25	0.0±0.7	59.5	131.2±16.7	44.5	49.9	139.6
Atenolol	508.0±50	0.0±1.4	81.9	261.1±30.2	74.3	45.0	236.7
Candesartan	754.8±70	0.0±2.7	115.4	410.3±35.7	122.5	59.6	310.5
Captopril	427.0±40	0.0±2.2	74.8	212.1±27.3	54.4	54.3	170.7
Carvedilol	655.2±55	0.0±2.1	101.4	350.1±31.5	119.6	53.9	325.1
Dipyridamole	806.5±75	0.0±3.0	122.9	441.5±37.1	139.4	81.6	373.0
Edoxaban	-	-	-	-	139.1	73.3	383.2
Quinapril	662.0±55	0.0±2.1	102.3	354.1±31.5	119.5	52.3	360.1
Ramipril	616.2±55	0.0±1.9	96.1	326.4±31.5	111.4	50.2	346.8
Rivaroxaban	732.6±60	0.0±2.4	106.9	396.9±32.9	106.7	61.0	298.5
Ticagrelor	777.6±70	0.0±2.8	118.7	424.0±35.7	126.3	63.3	311.9
Warfarin	515.2±50	0.0±1.4	82.9	188.8±23.6	84.4	58.7	235.8

$$H(G) = \sum_{r \in E(G)} \frac{2}{\Phi_r + \Phi_s}.$$

H-index is used to obtain the extremal values of chemical trees in terms of the order and dominance number of molecular graphs (Hu and Zhang (2022)).

4. Sum Connectivity Index

Zhou and Trinajstić introduced the sum connectivity index in 2010 (Zhou and Trinajstić (2010)). It was proposed while giving the general form of the H-index. It is defined as:

$$SCI(G) = \sum_{r \in E(G)} \frac{1}{\sqrt{\Phi_r + \Phi_s}}.$$

The SC-index is extremely important in calculating the correlation coefficient for boiling point ($r = 0.925$) and anti-tuberculosis medicine enthalpy (Adnan et al. (2022)).

5. First and Second Zagreb Index

In 1972, Gutman and Trinajstić published the first and second Zagreb indices (Gutman and Trinajstić (1972)). They found a relationship between total pi-electron energy and the Z-indices. These indices are determined as follows:

$$M_1(G) = \sum_{r \in E(G)} (\Phi_r + \Phi_s),$$

and

$$M_2(G) = \sum_{r \in E(G)} (\Phi_r \times \Phi_s).$$

These indices are employed in the investigation of the structures of connected quasi-unicyclic graphs (Aghel et al. (2019)). The Z-indices are extremely useful for understanding the properties of corona drugs (Wei et al. (2022)).

6. Hyper Zagreb Index

In 2013, Shirdel proposed the Hyper Zagreb index (Shirdel et al. (2013)). The mathematical statement of the HM-index is:

$$HM(G) = \sum_{r \in E(G)} (\Phi_r + \Phi_s)^2.$$

There are many criteria to check the Hamiltonianity of the connected graphs. The HM-index can describe the Hamiltonian nature of simple graphs like trees, wheels, and chemical graphs (Lu and Zhou (2022)). The HM-index has the ability to explore the properties of middle graphs that are helpful in the network theory (Bommanahal and Policepatil (2022)).

7. Reduced Second Zagreb Index

In 2013, Ranjini et al. introduced reduced Zagreb indices (Ranjini et al. (2013)). The RM_2 -index mathematical formula is as follows:

$$RM_2(G) = \sum_{r \in E(G)} (\Phi_r - 1)(\Phi_s - 1).$$

8. Forgotten Index

Furtula and Gutman introduced the Forgotten index to estimate the properties of carbon compounds in 2015 (Furtula and Gutman (2015)). They noticed the data set of isomers of octanes and observed that the F-index has a correlation with many properties of the octane isomers. It was observed that the F-predictive index's power is quite comparable to the first Zagreb index. Using M_1 and the F-index, entropy and the acentric factor both provide correlation values greater than 0.95. It is computed as follows:

$$F(G) = \sum_{r \in E(G)} (\Phi_r^2 + \Phi_s^2).$$

The F-index has a correlation with the physical-chemical properties of anti-tuberculosis drugs. Tuberculosis is a common lung disease caused by inhaling microscopic droplets from a sick person's cough or sneeze (Adnan et al. (2022)).

9. Hyper Forgotten Index

The Inspiring Forgotten Index proposed the hyper Forgotten Index, which is a vertex degree-based index. The HF-index is mathematically defined as follows:

$$HF(G) = \sum_{r \in E(G)} (\Phi_r^2 + \Phi_s^2)^2.$$

The hyper forgotten index can be used to describe the properties of nanostar dendrimers (Ghobadi et al. (2022)). Dendrimers are branching molecules that have a center component attached to tree-like branches that are repeatedly connected to each other. For more information about topological indices of chemicals, see (Mahboob et al. (2022), Alam et al. (2022)).

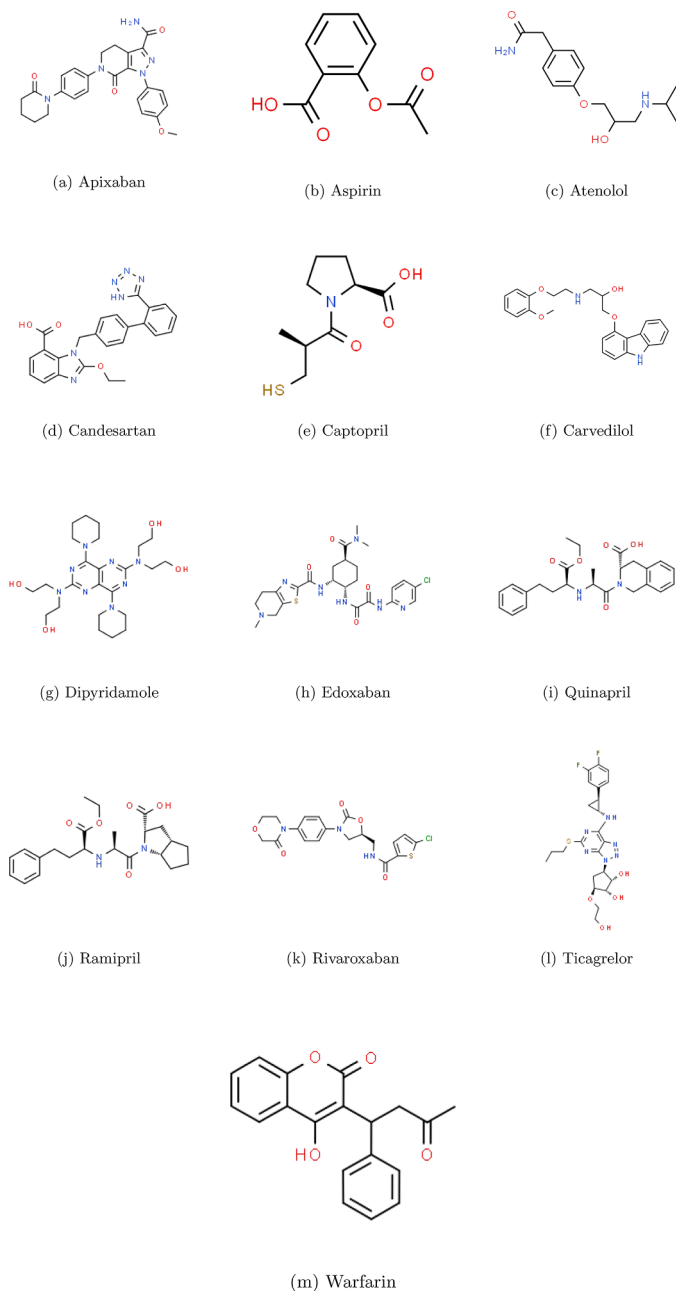
3. Methods and strategies

There are three tools used to determine the degree-related indices: edges partition, vertex partition, and degree counting. In order to investigate the topology of the relevant molecular graph of the underlying chemical, we employ graph theoretic approaches. First of all, count the degrees of all the vertices and classify the vertices into different groups. An edge is formed by the link between vertices. The edges are partitioned into five groups. Because all of the indices are simply vertex degree-based, the distance or eccentricity technique is useless here. The software chemSketch is used to draw the 2D structures of medicines. Correlation graphs or line graphs can be drawn with the help of Microsoft Excel. Other software like Mathematica and Maple can also be used for drawing line graphs. To calculate the values of topological

Table 2

Topological indices of heart attack drugs

Drug Names	RI(G)	RR(G)	H(G)	SCI(G)	M ₁ (G)	M ₂ (G)	HM(G)	RM ₂ (G)	F(G)	HF(G)
Apixaban	16.14	88.08	15.66	16.95	180	217	900	74	466	6388
Aspirin	6.18	28.17	5.83	6.14	59	63	273	17	147	1761
Atenolol	8.96	41.15	8.5	8.97	86	91	394	24	212	2446
Candesartan	13.47	87.07	15.9	17.09	177	211	869	71	447	5901
Captopril	6.57	31.69	6.2	6.55	66	76	324	24	172	2356
Carvedilol	14.75	75.98	9.33	15.43	114	180	738	59	378	4774
Dipyridamole	17.69	89.84	17.4	18.32	182	215	880	72	450	5840
Edoxaban	9.43	93.74	16.8	18.28	194	226	958	72	506	6718
Quinapril	15.40	77.91	14.86	15.84	160	185	774	59	384	4312
Ramipril	13.99	71.18	13.53	14.43	146	169	756	54	368	4754
Rivaroxaban	14.52	75.12	13.56	14.69	154	181	937	59	394	5172
Ticagrelor	17.40	95.84	16.86	18.28	196	238	988	82	512	7116
Warfarin	11.07	58.52	10.7	11.52	120	142	592	47	308	4144

**Fig. 1.** Structures of Heart Attack Drugs

indices given in Table 2, we utilized the calculator and verified the calculated results with the help of Matlab. For statistical calculations, SPSS is the best software.

4. Topological analysis of heart attack drugs structures

In this article, the 13 two-dimensional structures of heart attack drugs are analyzed with the help of 10 famous degree-based molecular descriptors. These structures are three-dimensional in reality, but topological indices discuss the two-dimensional graphs of medicines. These medicines are helpful for not only heart attacks but also for other heart-related disorders. Many drugs are used to treat heart disorders, but the 13 drugs that we studied are: apixaban, aspirin, atenolol, candesartan, captopril, carvedilol, dipyridamole, edoxaban, quinapril, ramipril, rivaroxaban, ticagrelor, and warfarin. Aspirin is the most important of all the medications. Aspirin works by preventing the synthesis of naturally occurring chemicals that cause fever, discomfort, stiffness, and blood clots. The structures of these drugs are given in Fig. 1.

The QSPR analysis of the above-mentioned topological indices is described, and it is demonstrated that these topological indices are strongly correlated with the physical features of chemical substances used to treat heart attacks. The experimental values of the seven physicochemical properties of heart attack medicines are shown in Table 1. The estimated values of these drugs are calculated with the help of degree-based descriptors. The correlation between estimated values and experimental values indicates the importance of topological indices. For the investigation of heart attack medications, seven experimental parameters are used. These parameters are shown in Table 1 as the boiling point, vapor pressure, enthalpy, flash point, molar refraction, surface tension, and molar volume. All of the two-dimensional structures and drug characteristics shown in Fig. 1 and Table 1 are taken from PubChem. Table 1 presents some parameters with sharp values and others without sharp values. These values may be found in a variety of ranges, including flash point, vapor pressure, and boiling point. Only sharp values are considered in the correlation coefficient calculations.

All the topological indices are determined by using the edge degree and formulae given above. The degrees of the vertices are first calculated and then classified based on the degrees of the edges. For the estimation of the properties of heart attack drugs, ten degree-based topological indices are computed: the randic index, the reciprocal randic index, the harmonic index, the sum connectivity index, the first Zagreb index, the second Zagreb index, the hyper Zagreb index, the reduced second Zagreb index, the forgotten index, and the hyper forgotten index. Table 2 shows that the HF-index has the greatest values and the H-index has the lowest values. The edges of all the structures are classified according to degree. There are five major types of edge bundles. The maximum degree of a vertex is three, and the minimum degree is one. The edge parcel ($e = (3, 2)$) has the most parcels. The values of all the degree-based indices are given in Table 2, and the five parcels of edges

are:

$$\begin{aligned} E_{(1,2)} &= \{e = rs \in E(G) | \Phi_r = 1, \Phi_s = 2\}, \\ E_{(1,3)} &= \{e = rs \in E(G) | \Phi_r = 1, \Phi_s = 3\}, \\ E_{(2,2)} &= \{e = rs \in E(G) | \Phi_r = 2, \Phi_s = 2\}, \\ E_{(3,2)} &= \{e = rs \in E(G) | \Phi_r = 3, \Phi_s = 2\}, \text{ and} \\ E_{(3,3)} &= \{e = rs \in E(G) | \Phi_r = 3, \Phi_s = 3\}. \end{aligned}$$

5. Regression models

In order to model the relationship between two variables, linear regression fits a linear equation to the observed data. The first variable is regarded as an explanatory variable, whereas the second is regarded as a dependent variable. The values of linear regression for all the topological indices can be calculated by the regression model as given below:

$$P = a + bX,$$

where P is the dependent variable and X is the independent parameter. The parameter P represents the physical or chemical properties of the heart attack medicine, and X stands for the estimated values of characteristics or topological indices. When $X = 0$, the constant, or intercept, parameter indicates the predicted response. The slope parameter b specifies the predicted increment in response per unit change in X.

1. Randic Index $R(G)$

$$\begin{aligned} BP &= 145.8865 + 37.1229[R(G)] \\ VP &= 0.4013 + 0.1312[R(G)] \\ EV &= 37.622 + 4.6339[R(G)] \\ FP &= 14.2423 + 23.9874[R(G)] \\ MR &= 23.3567 + 6.4267[R(G)] \\ ST &= 44.9907 + 1.0824[R(G)] \\ MV &= 106.5036 + 14.6811[R(G)] \end{aligned}$$

2. Reciprocal Randic Index $RR(G)$

$$\begin{aligned} BP &= 188.0066 + 6.448[RR(G)] \\ VP &= 0.5049 + 0.02345[RR(G)] \\ EV &= 42.6236 + 0.8086[RR(G)] \\ FP &= 43.2789 + 4.1398[RR(G)] \\ MR &= 13.689 + 1.3013[RR(G)] \\ ST &= 39.1171 + 0.2795[RR(G)] \\ MV &= 86.6523 + 2.9409[RR(G)] \end{aligned}$$

3. Harmonic Index $H(G)$

$$\begin{aligned} BP &= 196.1181 + 35.0137[H(G)] \\ VP &= 0.5112 + 0.129[H(G)] \\ EV &= 43.456 + 4.4059[H(G)] \\ FP &= 48.852 + 22.4505[H(G)] \\ MR &= 17.2629 + 6.9235[H(G)] \\ ST &= 37.5685 + 1.6696[H(G)] \\ MV &= 90.4781 + 15.9819[H(G)] \end{aligned}$$

4. Sum Connectivity Index $SC(G)$

$$\begin{aligned} BP &= 158.0807 + 34.4074[SC(G)] \\ VP &= 0.4126 + 0.1239[SC(G)] \\ EV &= 38.7217 + 4.3258[SC(G)] \\ FP &= 22.4437 + 22.2092[SC(G)] \\ MR &= 6.2393 + 7.0502[SC(G)] \\ ST &= 37.828 + 1.4923[SC(G)] \\ MV &= 66.3878 + 16.1776[SC(G)] \end{aligned}$$

5. First Zagreb Index $M_1(G)$

$$\begin{aligned} BP &= 202.5376 + 3.1198[M_1(G)] \\ VP &= 0.5379 + 0.01149[M_1(G)] \\ EV &= 44.5255 + 0.3907[M_1(G)] \\ FP &= 54.0473 + 1.9925[M_1(G)] \\ MR &= 19.6546 + 0.6064[M_1(G)] \\ ST &= 38.7798 + 0.1417[M_1(G)] \\ MV &= 99.5275 + 1.3748[M_1(G)] \end{aligned}$$

6. Second Zagreb Index $M_2(G)$

$$\begin{aligned} BP &= 211.1611 + 2.5473[M_2(G)] \\ VP &= 0.5681 + 0.009392[M_2(G)] \\ EV &= 45.5245 + 0.3195[M_2(G)] \\ FP &= 59.0609 + 1.6299[M_2(G)] \\ MR &= 18.799 + 18.799[M_2(G)] \\ ST &= 39.89 + 0.1119[M_2(G)] \\ MV &= 100.3789 + 1.1442[M_2(G)] \end{aligned}$$

7. Hyper Zagreb Index $HM(G)$

$$\begin{aligned} BP &= 206.8944 + 0.6011[HM(G)] \\ VP &= 0.5837 + 0.002172[HM(G)] \\ EV &= 45.8131 + 0.07421[HM(G)] \\ FP &= 54.9513 + 0.3866[HM(G)] \\ MR &= 20.2882 + 0.1177[HM(G)] \\ ST &= 40.4079 + 0.02545[HM(G)] \\ MV &= 100.8191 + 0.2669[HM(G)] \end{aligned}$$

8. Redefined Second Zagreb Index $RM_2(G)$

$$\begin{aligned} BP &= 260.7616 + 6.8814[RM_2(G)] \\ VP &= 0.7175 + 0.026[RM_2(G)] \\ EV &= 51.6492 + 0.8648[RM_2(G)] \\ FP &= 92.0211 + 4.3801[RM_2(G)] \\ MR &= 28.8438 + 1.3904[RM_2(G)] \\ ST &= 41.6148 + 0.3125[RM_2(G)] \\ MV &= 126.2613 + 3.0447[RM_2(G)] \end{aligned}$$

9. Forgotten Index $F(G)$

$$\begin{aligned} BP &= 186.5784 + 1.2525[F(G)] \\ VP &= 0.4847 + 0.004597[F(G)] \\ EV &= 42.5825 + 0.1567[F(G)] \\ FP &= 43.256 + 0.8016[F(G)] \\ MR &= 15.7638 + 0.2451[F(G)] \\ ST &= 38.9238 + 0.0544[F(G)] \\ MV &= 93.621 + 0.5477[F(G)] \end{aligned}$$

10. Hyper Forgotten Index $HF(G)$

$$\begin{aligned} BP &= 231.7299 + 0.08672[HF(G)] \\ VP &= 0.5984 + 0.0003296[HF(G)] \\ EV &= 48.2263 + 0.01085[HF(G)] \\ FP &= 74.2687 + 0.05504[HF(G)] \\ MR &= 26.2362 + 0.01664[HF(G)] \\ ST &= 39.8113 + 0.003997[HF(G)] \\ MV &= 123.292 + 0.03587[HF(G)] \end{aligned}$$

5.1. Calculations of statistical parameters

There are several statistical parameters used in this section. A parameter is a valuable statistical analysis component that relates to the qualities that are used to characterize a certain population. It is used to

Table 3

Statistical data for Randic index.

Properties	N	a	b	r	r^2	F	p	Indicator
BP	12	145.8865	37.1229	0.944	0.8912	81.9197	0.000	Significant
VP	12	0.4013	0.1312	0.7691	0.5915	14.48	0.003	Significant
EV	12	37.622	4.6339	0.9392	0.8822	74.8556	0.000	Significant
FP	12	14.2423	23.9874	0.9097	0.8276	48.0094	0.000	Significant
MR	13	23.3567	6.4267	0.8084	0.6535	20.7447	0.000	Significant
ST	13	44.9907	1.0824	0.4247	0.1804	2.4204	0.148	Not Significant
MV	13	106.5036	14.6811	0.7539	0.5684	14.4845	0.002	Significant

Table 4

Statistical data for reciprocal Randic index

Properties	N	a	b	r	r^2	F	p	Indicator
BP	12	188.0066	6.448	0.9664	0.934	141.422	0.000	Significant
VP	12	0.5049	0.02345	0.8102	0.6563	19.0992	0.001	Significant
EV	12	42.6236	0.8086	0.966	0.9331	139.4598	0.000	Significant
FP	12	43.2789	4.1398	0.9253	0.8563	59.5733	0.000	Significant
MR	13	13.689	1.3013	0.9784	0.9573	246.586	0.000	Significant
ST	13	39.1171	0.2795	0.6556	0.4298	8.2905	0.014	Significant
MV	13	86.6523	2.9409	0.9027	0.8149	48.4293	0.000	Significant

Table 5

Statistical data for harmonic index

Properties	N	a	b	r	r^2	F	p	Indicator
BP	12	196.1181	35.0137	0.9263	0.858	60.4381	0.000	Significant
VP	12	0.5112	0.1292	0.788	0.6209	16.3811	0.002	Significant
EV	12	43.456	4.4059	0.929	0.8631	63.0375	0.000	Significant
FP	12	48.852	22.4505	0.8858	0.7846	36.4228	0.000	Significant
MR	13	17.2629	6.9235	0.9182	0.8431	59.0918	0.000	Significant
ST	13	37.5685	1.6696	0.6906	0.477	10.0319	0.008	Significant
MV	13	90.4781	15.9819	0.8653	0.7487	32.7712	0.000	Significant

Table 6

Statistical data for sum connectivity index

Properties	N	a	b	r	r^2	F	p	Indicator
BP	12	158.0807	34.4074	0.9657	0.9326	138.4437	0.000	Significant
VP	12	0.4126	0.1239	0.8018	0.6428	17.9983	0.001	Significant
EV	12	38.7217	4.3258	0.9677	0.9365	147.4209	0.000	Significant
FP	12	22.4437	22.2092	0.9297	0.8643	63.6654	0.000	Significant
MR	13	6.2393	7.0502	0.9898	0.9798	533.2602	0.000	Significant
ST	13	37.828	1.4923	0.6535	0.4271	8.2	0.015	Significant
MV	13	66.3878	16.1776	0.9273	0.859	67.4682	0.000	Significant

Table 7

Statistical data for first zagreb index

Properties	N	a	b	r	r^2	F	p	Indicator
BP	12	202.5376	3.1198	0.9464	0.8957	85.8576	0.000	Significant
VP	12	0.5379	0.01149	0.8035	0.6456	18.22	0.001	Significant
EV	12	44.5255	0.3907	0.9446	0.8922	82.7494	0.000	Significant
FP	12	54.0473	1.9925	0.9014	0.8126	43.3503	0.000	Significant
MR	13	19.6546	0.6064	0.9332	0.8708	74.138	0.000	Significant
ST	13	38.7798	0.1417	0.6803	0.4628	9.478	0.010	Significant
MV	13	99.5275	1.3748	0.8637	0.7459	32.296	0.000	Significant

Table 8

Statistical data for second zagreb index.

Properties	N	a	b	r	r^2	F	p	Indicator
BP	12	211.1611	2.5473	0.9627	0.9267	126.423	0.000	Significant
VP	12	0.5681	0.009392	0.8182	0.6694	20.2457	0.001	Significant
EV	12	45.5245	0.3195	0.9623	0.9259	125.0214	0.000	Significant
FP	12	59.0609	1.6299	0.9186	0.8438	54.0271	0.000	Significant
MR	13	18.799	0.512	0.968	0.9371	163.82	0.000	Significant
ST	13	39.89	0.1119	0.66	0.4355	8.4877	0.014	Significant
MV	13	100.3789	1.1442	0.8832	0.78	39.0003	0.000	Significant

Table 9

Statistical data for hyper zagreb index

Properties	N	a	b	r	r^2	F	p	Indicator
BP	12	206.8944	0.6011	0.962	0.9255	124.2084	0.000	Significant
VP	12	0.5837	0.002172	0.8012	0.6419	17.9233	0.001	Significant
EV	12	45.8131	0.07421	0.9467	0.8962	86.3133	0.000	Significant
FP	12	54.9513	0.3866	0.9227	0.8513	57.2713	0.000	Significant
MR	13	20.2882	0.1177	0.94	0.8837	83.5696	0.000	Significant
ST	13	40.4079	0.02545	0.6342	0.4023	7.4025	0.019	Significant
MV	13	100.8191	0.2669	0.8707	0.7581	34.4757	0.000	Significant

Table 10

Statistical data for redefined second zagreb index

Properties	N	a	b	r	r^2	F	p	Indicator
BP	12	260.7616	6.8814	0.9532	0.9086	99.4046	0.000	Significant
VP	12	0.7175	0.026	0.8301	0.6891	22.1679	0.000	Significant
EV	12	51.6492	0.8648	0.9548	0.9116	103.1529	0.000	Significant
FP	12	92.0211	4.3801	0.9049	0.8188	45.1754	0.000	Significant
MR	13	28.8438	1.3904	0.9495	0.9016	100.7858	0.000	Significant
ST	13	41.6148	0.3125	0.6656	0.443	8.7495	0.013	Significant
MV	13	126.2613	3.0447	0.8489	0.7206	28.3655	0.000	Significant

Table 11

Statistical data for forgotten index

Properties	N	a	b	r	r^2	F	p	Indicator
BP	12	186.5784	1.2525	0.9635	0.9283	129.5143	0.000	Significant
VP	12	0.4847	0.004597	0.8153	0.6646	19.8191	0.001	Significant
EV	12	42.5825	0.1567	0.9607	0.9229	119.6193	0.000	Significant
FP	12	43.256	0.8016	0.9196	0.8457	54.8293	0.000	Significant
MR	13	15.7638	0.2451	0.962	0.9254	136.476	0.000	Significant
ST	13	38.9238	0.0544	0.666	0.4435	8.7681	0.012	Significant
MV	13	93.621	0.5477	0.8775	0.7701	36.8452	0.000	Significant

express a specific property of the entire population. The term "N" is the quantity of members in a sample. The values of N are 12 or 13 as given in Tables 3–12. The term "a" is the constant, and "b" is the slope value. A correlation coefficient r is a statistical indicator of how well changes in one variable's value indicate changes in another. The value of r may be positive or negative. The positive value shows the direct relation between two things, and the negative value shows an inverse relation. All the correlation coefficients have positive values, so they have a strongly direct relationship. The range of correlation coefficients is $-1 \leq r \leq 1$. A p-value calculates the chance of obtaining the outcomes that were observed, supposing that the null hypothesis is correct. All the values of p in this article are greater than or equal to 0.05, and all outcomes are significant. The correlation of all properties is good, except for surface tension. The greater the F-value, the greater the variance between sample averages and within-sample variation. The lower value of p means the higher value of F.

Table 3 shows the relationship between the R-index and the seven heart attack medicine characteristics. The R-index has a maximum and minimum correlation of 0.944 and 0.4247, respectively. There is just one insignificant number in the whole computation performed in this article, and it is found in Table 3. All the parameters in Table 4 show the significant values of p. Table 5 presents the numerical results for the H-

index. The correlation coefficients for the H-index vary from 0.9263 to 0.6906. Table 6 presents the statistical parameter values associated with the SC-index. All the correlation coefficients for the SC-index are greater than 0.8, which is an excellent experimental result. The characteristics of the anti-heart attack medicines shown in Tables 7 and 8 were estimated extremely well using the standard first and second Z-indices. The term r^2 is used for the absolute values of r. According to the inverse and direct relationship between two parameters, r values might be either negative or positive. The values N in Table 9 are 12 and 13. In Table 9, the F-values for the hyper-Zagreb index vary from 7.4025 to 124.2084. Information regarding the redefined Zagreb index is shown in Table 10. The correlation for this index ranges from 0.6656 to 0.9548. Table 11 presents the statistical variables associated with the forgotten index. For this index, every p value is less than 0.01 and every F value is more than 8.7681. For predicting the properties of heart attack medications, the HF index has been proven to be more accurate than the F-index. According to Table 12, the HF index has significant values with a p value less than 0.09. Strong positive correlations exist between this index and all experimental values. The correlation for the HF index is between 0.689 and 0.9401. The parameters in Table 12 with correlation factors above 0.9 are BP (0.9401), EV (0.9374), and MR (0.9198).

Table 12

Statistical data for hyper forgotten index

Properties	N	a	b	r	r^2	F	p	Indicator
BP	12	231.7299	0.08672	0.9401	0.8838	76.0491	0.000	Significant
VP	12	0.5984	0.0003296	0.8238	0.6787	21.1216	0.000	Significant
EV	12	48.2263	0.01085	0.9374	0.8787	72.4635	0.000	Significant
FP	12	74.2687	0.05504	0.8898	0.7918	38.038	0.000	Significant
MR	13	26.2362	0.01664	0.9198	0.846	60.4218	0.000	Significant
ST	13	39.8113	0.003997	0.689	0.4747	9.9398	0.009	Significant
MV	13	123.292	0.03587	0.8092	0.6549	20.871	0.000	Significant

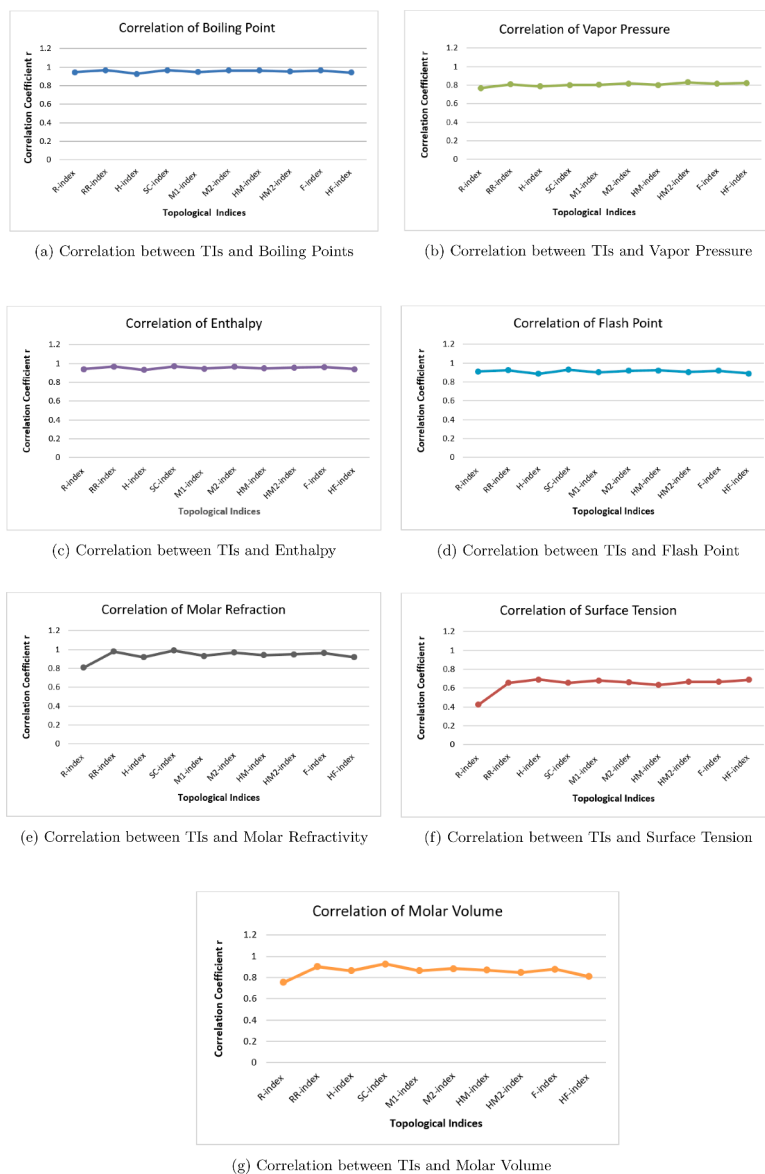


Fig. 2. Correlation between TIs and all Physical and chemical properties

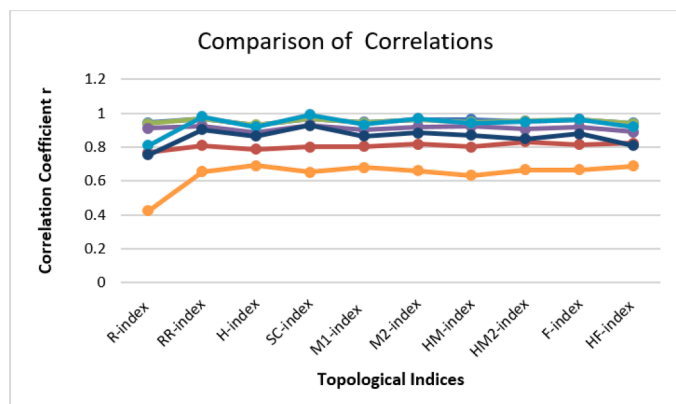


Fig. 3. Comparison of all correlation coefficient

6. Graphical analysis of correlation coefficients

A graph is a very powerful visual tool because it presents data quickly, allows for comparison, and may indicate patterns and relationships within the data such as changes over time, frequency distribution, correlation, or relative proportion of a whole. There are different types of graphs according to the nature of the data, such as bar charts, line graphs, area graphs, scatter plots, pie charts, pictographs, column charts, and bubble charts. There are seven line graphs representing the correlation between the properties and topological indices given in Fig. 2. The comparison of all the correlations is given in Fig. 3.

A line graph, also known as a line plot or a line chart, is a graph that connects individual data points with lines. The line graphs are drawn with different colors to make a difference between the correlations of the different parameters. The names of the topological indices are displayed on the x-axis, while the correlation values are displayed on the y-axis. The boiling point correlation graph is blue in color and has a value between 0.8 and 1.0, indicating a very good correlation. All of the graphs have a positive x-axis since there is a direct relationship between topological indices (estimated values) and properties (experimental values). The surface tension correlation coefficient graph is drawn in brown and

Table 13

Comparison of correlation coefficients for topological indices

Indices	BP	VP	EV	FP	MR	ST	MV
R(G)	0.944	0.7691	0.9392	0.9097	0.8084	0.4247	0.7539
RR(G)	0.9664	0.8102	0.966	0.9253	0.9784	0.6556	0.9027
H(G)	0.9263	0.788	0.929	0.8858	0.9182	0.6906	0.8653
SC(G)	0.9657	0.8018	0.9677	0.9297	0.9898	0.6535	0.9273
$M_1(G)$	0.9464	0.8035	0.9446	0.9014	0.9332	0.6803	0.8637
$M_2(G)$	0.9627	0.8182	0.9623	0.9186	0.968	0.66	0.8832
HM(G)	0.962	0.8012	0.9467	0.9227	0.94	0.6342	0.8707
$RM_2(G)$	0.9532	0.8301	0.9548	0.9049	0.9495	0.6656	0.8489
F(G)	0.9635	0.8153	0.9607	0.9196	0.962	0.666	0.8775
HF(G)	0.9401	0.8238	0.9374	0.8898	0.9198	0.689	0.8092

ranges from 0.4 to 0.8, indicating a moderate association. The surface tension graph is moderate, but all of the other graphs in Fig. 2 indicate incredibly strong correlation values for all topological indices.

These line graphs were drawn with the help of Microsoft Excel. The scatter plot is also useful for analyzing the correlation coefficients. The values of all the correlation coefficients are given in Table 13. The variation in the values of the correlation coefficients can be noted by the numerical values in Table 13. The surface tension has the least amount of correlation with the degree-related indices. All the other properties have excellent correlations with molecular indices of structures. The 2D graphs can be drawn by using the table of numerical data. Graphs give an easy and clear way to understand the variation and comparison of data. The properties like boiling point, vapor pressure, enthalpy, and flash point are easily determined with the help of the indices mentioned above because the correlation of all these properties is almost above 0.9.

7. Applications of topological indices

Topological indices have many applications in many fields of science, particularly chemistry, pharmacy, and network theory. The first molecular descriptor was a distance-based index proposed by a chemist named Wiener for estimating the characteristics of the different classes of organic compounds. By inspiring his work, degree-based indices were introduced that have many applications in chemistry, like the Randic index, which is used to determine the branching structures of carbon compounds, and the Zagreb indices, which are used to understand the tree's structures. All the indices defined above are applicable to describing the physical features of heart attack medicines. There are seven properties of drugs that are examined with the help of indices.

The reciprocal index is used to calculate the boiling points of drugs, and a correlation coefficient shows $r = 0.9664$. The surface tension of drugs can be determined by the harmonic index because it shows the highest correlation value with ST ($r = 0.6906$). This correlation value is not particularly good, but it is significant for calculating surface tension. Enthalpy (0.9677), flash point (0.9297), molar refractivity (0.9297), and molar volume (0.9273) are determined by using the sum connectivity index. This index is the most efficient index because the relationship between its four properties has the highest values. The redefined second Zagreb index is used to find the estimated values of vapor pressure ($r = 0.8301$).

8. Conclusion and discussion

The medications used to treat heart attacks are examined in the current work, and several degree-related numerical descriptors are used for each drug. There are a total of 13 heart attack drugs studied with the help of 10 degree-based indices. Table 2 and Table 13 and Fig. 2 show the values of degree-based topological indices and how they correlate. Any new medicine must have certain structural characteristics, which may be found by QSPR modeling with TIs. The objective of this paper is to quickly and cheaply extract information about the topology of a structure by utilizing topological indices. A number of conclusions may

be drawn from the data in Tables 3–13 for the stated degree-related topological indices.

The Randic index is a widely used structural and molecular descriptor. Table 13 makes it simple to see that the Randic index demonstrates high correlation with all of the physical characteristics of heart attack medications, with the exception of their surface tension, with a correlation coefficient value of $r = 0.4247$. Moreover, we can observe that the correlation coefficient ranges from 0.7539 to 0.944. The QSPR analysis of the reciprocal randic (RR) index demonstrates that the RR-index, with a correlation coefficient value of $r = 0.9664$, is a good predictor for the boiling points of medicines.

The sum connectivity index shows a very high correlation with enthalpy (0.9677), flash point (0.9297), molar refractivity (0.9898), and molar volume (0.9273) of the heart attack drugs. It is the most efficient index because it gives the highest correlation with four physical-chemical features of drugs. It is clear from Tables 7 and 8 that the Zagreb indices show a good correlation with all properties except surface tension (0.6803, 0.66). The correlation ranges for the Zagreb indices are from 0.8035 to 0.968. Furthermore, QSPR research on the second Zagreb index shows that it is more predictive than the Randic index. The highest correlation exists between vapour pressure and the RM_2 index. Except for surface tension, the forgotten index and hyper forgotten index have excellent correlations with all properties, but they do not have the highest correlation values with any property.

These indices can be used as predictive factors in QSPR investigations. The physical-chemical properties of drugs used to treat heart attacks are investigated in order to assess the predictive power of these indices. The obtained outcomes show that all of the results are extremely relevant and that there is a strong positive linear relationship between the molecular volume, boiling point, enthalpy, flash point, molar refractivity, vapor pressure, and TIs.

8.1. Limitations

A huge number of different types of topological indices have been introduced for gaining information about chemical graphs and computer networks, but these are still insufficient for computing all of the parameters of compounds. As a result, additional molecular descriptors are required for a more theoretical and simple method of determining the characteristics of chemical structures, particularly heart attack medications.

8.2. Future work

Similarly, we have plan to investigate the chemical and physical characteristics of medicines about treatment of lung cancer, diabetes and few other serious diseases which can be analyzed with newly introduced index comparatively producing better results over chemical graphs. Our second motive is working on 3 dimensional and geometrical descriptors in drugs.

8.3. Data Availability

In this article, no data were utilized.

8.4. Funding statement

This research received no funding.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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